

THE UNIVERSITY OF NEW SOUTH WALES
SCHOOL OF PHYSICS

MIDSESSION EXAMINATION

PHYS3114 – Electrodynamics

Term 3, 2021

1. Time allowed – 50 minutes {+ 15 min for download/reading + 30 min for capture/upload}
2. Total number of questions – 2
3. Total marks available – 28 (pro rata to 15% of course assessment)
4. You are allowed to use your textbook and notes during the exam.
5. Prepare a single file (pdf files preferred) with all of your responses to upload to Moodle by the end of the allotted exam time.
6. Additional time will be allocated for submitting your answers
7. Your submitted file should have your name and zID on the first page of your submitted file (preferably in the header or footer on every page).
8. When uploading please use the following naming convention: PHYS3114 zID SURNAME NAME (e.g., PHYS3114 z5556789 Riddle Tom). NOTE: failure to name your exam file correctly may result in marking delays if your file is misplaced.
9. **Candidates must verify the upload is successful and contact the exam convener immediately if there is an issue.**
10. Answer all questions concisely and legibly.
11. In some cases, sketching your answer can be helpful.
12. Questions are not all worth the same marks; questions are worth the marks shown.
13. You may use a university-approved calculator.

PHYS3114 — Definitions and Formulae

$$F(s) = \int_{-\infty}^{+\infty} f(x) e^{-2\pi i x s} dx \quad (1)$$

$$\mathcal{F}\{f(x)\} = 2 \int_0^\infty f_e(x) \cos(2\pi s x) dx - 2i \int_0^\infty f_o(x) \sin(2\pi s x) dx \quad (2)$$

$$V(x', y') = C_1 C_2 \lambda^2 f_o^2 \iint_{\text{bfp}} F\left(\frac{\zeta}{\lambda f_o}, \frac{\eta}{\lambda f_o}\right) e^{-i2\pi(\frac{f_g}{D} x' \frac{\zeta}{\lambda f_o} + \frac{f_o}{D} y' \frac{\eta}{\lambda f_o})} d\left(\frac{\zeta}{\lambda f_o}\right) d\left(\frac{\eta}{\lambda f_o}\right) \quad (3)$$

$$\gamma(P_1, P_2) = \frac{\iint_{\text{source}} I(\zeta, \eta) e^{\frac{-2\pi i}{\lambda}(p\zeta + q\eta)} d\zeta d\eta}{\iint_{\text{source}} I(\zeta, \eta) d\zeta d\eta} \quad (4)$$

$$F(\rho) = 2\pi \int_0^\infty f(r) J_0(2\pi \rho r) r dr \quad (5)$$

$$\frac{m\lambda}{2} = \frac{r_m^2}{2} \left(\frac{1}{r_0} + \frac{1}{b_0} \right) \quad (6)$$

$$\frac{d^2 P}{dt^2} + \gamma \frac{dP}{dt} + \omega_0^2 P = \frac{e^2 N}{m} \left(E + \frac{\nu P}{\epsilon_0} \right). \quad (7)$$

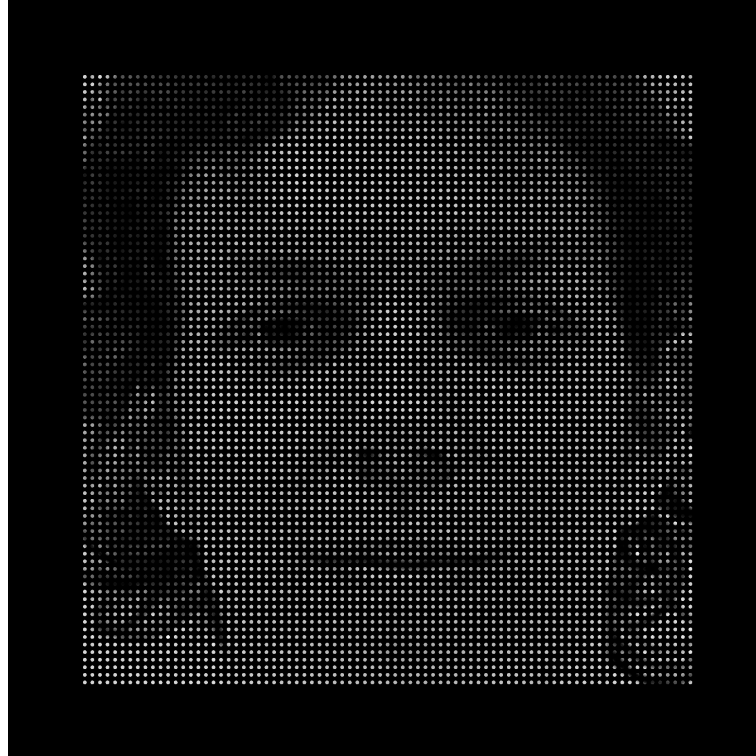
$$\hat{K} - 1 = \frac{\omega_p^2(\omega_{0'}^2 - \omega^2)}{(\omega_{0'}^2 - \omega^2)^2 + (\gamma\omega)^2} + i \frac{\omega_p^2 \omega \gamma}{(\omega_{0'}^2 - \omega^2)^2 + (\gamma\omega)^2} \quad (8)$$

Theorem	$f(x)$	$F(s)$
Similarity	$f(ax)$	$\frac{1}{ a }F\left(\frac{s}{a}\right)$
Linearity	$\alpha \cdot f(x) + \beta \cdot g(x)$	$\alpha \cdot F(s) + \beta \cdot G(s)$
Shift	$f(x - a)$	$e^{-2\pi ias}F(s)$
Convolution	$f(x) * g(x)$	$F(s)G(s)$
Derivative	$\frac{d}{dx}(f(x))$	$2\pi isF(s)$
Rayleigh	$\int_{-\infty}^{+\infty} f(x) ^2 dx$	$\int_{-\infty}^{+\infty} F(s) ^2 ds$

Table 1: Basic theorems of Fourier theory

Question 1 (12 marks)

Consider the following object to be used for Fourier filtering. It consists of a “familiar” face (image size, b) sampled with a 2D square array of delta function-like points that are separated by a distance, a . The object is placed in the object plane of an optical diffractometer and illuminated with a uniform plane wave of wavelength, λ . Filtering occurs in the back focal plane of a lens of focal length f_0 . The diffractometer has a magnification of 1.



- (a) Construct an expression for the object, $f(x, y)$, in terms of common Fourier functions.
- (b) With the aid of appropriate Fourier theorems, derive an expression for the diffracted field in terms of spatial dimensions, $F(\zeta, \eta)$, observed at the back focal plane of the transform lens.
- (c) A circular aperture is to be used as a low pass filter. With justification, determine the size of the aperture required to remove the array points, but retain all other spatial frequencies contained in the image.
- (d) Two such circular apertures are placed over adjacent diffraction peaks in Fourier plane and the filtered field is allowed to propagate through to the image plane. Derive an expression for the resulting filtered image and comment on how the image has been modified.

Question 2 (16 marks)

You are to assess two 1D aperture functions for their suitability for analysing spectral properties of an incident light source. The aperture functions are placed next to a lens with a focal length of 25 cm and the resulting diffraction pattern is observed in the back focal plane. Each aperture function has a finite width corresponding to 1cm. The apertures are given by:

$$\begin{aligned}\text{Aperture 1 : } f(x) &= \frac{1}{2}[1 - \cos(ax)] \\ \text{Aperture 2 : } f(x) &= \exp(ix/D)\end{aligned}\tag{9}$$

For each aperture:

- (a) Calculate the corresponding diffraction pattern observed at the back focal plane.
- (b) Derive an expression for the expected spectral resolution.
- (c) A sodium lamp consisting of two closely spaced spectral lines (589.6 nm and 589.0nm) is used to test the spectral resolution. Determine what value of a (and D) are needed to resolved the lines?
- (d) Calculate the diffracted intensity in each diffraction peak and discuss the implication for the intended spectroscopy application.